

Perturbation Methods (MATH 2015-1)

Spring 2026

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Homework March 4th, 2026, Due March 25th 10 : 00pm

Use the matched asymptotic method to provide a composite solution, accurate to leading order plus one correction, of the following problem

$$\varepsilon y''(x) - 2(x^2 + 1)y'(x) + 1 = 0$$

for $x \in [0, 1]$ with $y(0) = y(1) = 0$, and where $\varepsilon \ll 1$.

1. Develop asymptotic solutions. Create a composite solution.
2. For a given choice of parameters, compare your approximate solution with an accurate numerical solution.